Anirudh Sivarajan

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Education

University Of California, San Diego — La Jolla, CA
Master of Science in Bioengineering — *August 2023 – Present*

Purdue University — West Lafayette, IN
Bachelor of Science in Biomedical Engineering — *August 2019 – December 2022*

Experience

Forward Deployed AI Engineer — January 2026 – Present
Retina Robotics — *Greenwood, IN*

- Contributed to a warehouse automation project developing a computer vision model to detect and localize packages, improving overall model accuracy by 5%
- Preprocessed and audited a dataset of 15,000 images, correcting mislabeled bounding boxes and improving annotation accuracy by 15–20%
- Designed an automated Quality Control pipeline for validating box labels, reducing mislabeled images by 20%
- Deployed the Warehouse Automation Module (WAM) across 3- 5 client sites, traveled onsite to diagnose mechanical alignment and stability issues, and validated system performance in live environments
- Integrated camera/sensor hardware with the trained detection model, resolving mounting alignment and vibration-related stability issues to enable reliable real-time package localization and decrease mislabeled packages by 15-20%
- Managed the end-to-end deployment lifecycle, coordinating with hardware, ML, and client-side teams to reduce average site turnaround time by 10%

Applied Machine Learning Research Scientist — April 2024 – January 2026
University of California, San Diego — *La Jolla, CA*

- Built a biologically constrained RNN in PyTorch trained via a batched AUROC surrogate loss, combining pairwise logistic margin, spectral penalty via power iteration, and off-window suppression, to learn synaptic scaling factors over a Drosophila connectome
- Designed a composite loss function with four weighted components (discrimination, compactness, stability, suppression) and a curriculum warmup schedule, optimizing only biologically meaningful parameters (per-cell-type scaling factors, global scale)
- Trained a downstream SNN simulation pipeline in Python using multiprocessing across 50 seeds, transferring learned RNN connectivity weights to spiking dynamics

Research Asssociate I — August 2023 – April 2024
University of California, San Diego — *La Jolla, CA*

- Processed and analyzed 16S rRNA metagenomic datasets (50–100+ samples) using QIITA, QIIME2, and NGS pipelines, identifying over 90 microbial taxa across 50 operational taxonomic units (OTUs) and uncovering microbiome–drug interaction patterns
- Applied diversity and differential abundance analyses to characterize microbiome shifts associated with Alzheimer's disease, identifying significant alterations across multiple bacterial genera linked to disease progression
- Quantified microbiome composition changes in Alzheimer's cohorts, detecting over 100 significantly altered microbial species between patients and healthy controls through metagenomic analysis

Research Scientist Intern — March 2022 – August 2022
Northwestern University — *Evanston, IL*

- Analyzed single-cell RNA sequencing data from 10,000+ mouse kidney cells to investigate gene expression patterns driving kidney development and cellular differentiation
- Improved clustering accuracy by 15% with Seurat, enhancing cell-type resolution and enabling novel marker identification

Technical Skills

Languages: Python, C/C++, R, SQL, MATLAB, Power BI, French, Tamil
Libraries: Pandas, NumPy, Matplotlib, TensorFlow, PyTorch, Scikit-learn, SciPy, Seaborn, Statsmodels, Brian2
AI Skills/Tools: Ultralytics YOLO, RT-DETR, OpenCV, MediaPipe, CVAT, ByteTrack, DeepSORT, Detectron2
Certifications: IBM AI Engineering(2026), Applications in Data Science(2022)